Capacity-Constrained Clustering for Auditable Redistricting Construction

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Abstract

This draft introduces T.15, a capacity-constrained clustering construction family for BISECT. The implementation emphasizes deterministic seeding, population-capacity status, repair-aware summaries, and RPLAN audit sidecars. The paper frames clustering as a construction alternative with explicit failure and repair states, not as a claim of global optimality, legal compliance, community preservation, or political fairness.

1 Introduction

Capacity-constrained clustering gives BISECT a construction family whose primary abstraction is assignment to population-capacity clusters. This makes capacity status, repair, and infeasibility visible rather than hidden inside a single final assignment.

2 Method

The staged method uses deterministic farthest-point seeds, assigns units subject to population capacity, and records whether the resulting assignment is valid, repaired, or infeasible. Small repair routines are treated as auditable post-processing rather than invisible cleanup.

The atlas view makes the key decision inspectable: a geographically nearest cluster is only a candidate, not an entitlement. If assigning a unit would overfill that cluster, the candidate is rejected with visible capacity arithmetic and the next eligible cluster is tried. The paper's artifact claim therefore depends on preserving seed order, rejected capacity candidates, fallback assignments, and final status.

2.1 Reproducible Procedure

The constructor operates on the graph and unit attributes supplied by the BISECT state pipeline. The current slice is graph-based and capacity-oriented: community-character weights and richer attribute clustering are future or optional inputs rather than claims of this baseline.

1. Select deterministic farthest-point seeds from the available units, breaking ties by stable unit identifier.
2. Assign remaining units to candidate clusters in deterministic order while respecting target population capacity and declared slack.
3. Mark the assignment `valid` when all declared constraints pass.
4. Run bounded repair only when the initial assignment is close enough to a valid profile for auditable post-processing.
5. Mark the result `repaired` when repair succeeds, `infeasible` when capacity cannot be satisfied, and `invalid` when validation fails after construction.

Status	Meaning	Artifact behavior
`valid`	Assignment satisfies declared profile	Write final sidecars
`repaired`	Bounded repair changed assignment	Write repair lineage
`infeasible`	Capacity cannot be satisfied	Write witness summary
`invalid`	Validation fails after construction	Block final export

Table 1: Capacity-clustering status taxonomy. Repair is visible lineage, not hidden cleanup.

3 Implementation

The implementation lives in `bisect-clustering` and is exposed through `bisect state --structure capacity-clustering` and YAML `structure: capacity-clustering`. Summaries record capacity status, repair status, parameter hashes, and algorithm lineage.

4 Audit Artifacts

Capacity-clustering outputs are framed as final-plan candidates only after the RPLAN/RCTX audit sidecars record assignments, capacity status, repair status, parameter hashes, and algorithm lineage. The verifier checks the declared audit profile; it does not certify that clustering choices satisfy fairness or legal requirements beyond the profile.

The sidecar contract binds the exported assignment to the constructor family, declared parameters, capacity and repair status, validation result, and algorithm lineage. Parameter hashes are used to detect drift between the reported run configuration and the artifact under verification. If repair changes assignments, that status remains part of the lineage rather than being collapsed into an ordinary valid plan.

5 Evaluation Plan

The current evidence layer includes deterministic seed tests, infeasible capacity fixtures, connectedness checks, RPLAN audit package tests, and a benchmark-tier RPLAN package generated from the capacity constructor on a 100-unit synthetic path. The next empirical layer should measure feasibility and quality on selected real-data smoke states.

Claim	Evidence now	Missing evidence	Status
Deterministic seeds	L0 seed fixtures	None for fixtures	Current
Capacity status reporting	L0 infeasibility tests	Broader case sweep	Current
RPLAN repair lineage	L1 sidecar tests	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke
Plan quality vs. baselines	None in this paper	Metrics and datasets	Future

Table 2: Evidence ladder for T.15 claims. Feasibility and quality are tracked separately.

Constructor	Capacity behavior	Failure artifact	Current role
Capacity clustering	explicit status/repair	witness or repair lineage	baseline
Spectral	balance via sweep	validation failure	comparison
Regionalization	capacity during merges	merge/status summary	comparison
Flow construction	capacity flow status	infeasibility witness	comparison
METIS	external partitioning	manifest/validation result	comparison

Table 3: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

6 Limitations

The current slice is not a full clustering benchmark suite. It establishes the artifact contract and deterministic baseline first; broader claims about quality, compactness, or communities of interest require additional sweeps.

Capacity feasibility is also not a normative districting claim. A valid or repaired output can still be weak on compactness, community preservation, partisan symmetry, or legal criteria outside the declared audit profile. Community-character weights are treated as future or optional evidence, not as part of the baseline claim.

7 Conclusion

T.15 adds clustering as a first-class construction family while preserving the audit fixed point: final plans must converge into RPLAN, RCTX, audit certificates, and manifests. The contribution is the inspectable capacity and repair contract, not a claim that clustered plans are globally optimal, politically fair, or legally sufficient.

References

BISECT Project. T.15 capacity-constrained clustering implementation spec, 2026. BISECT internal specification.